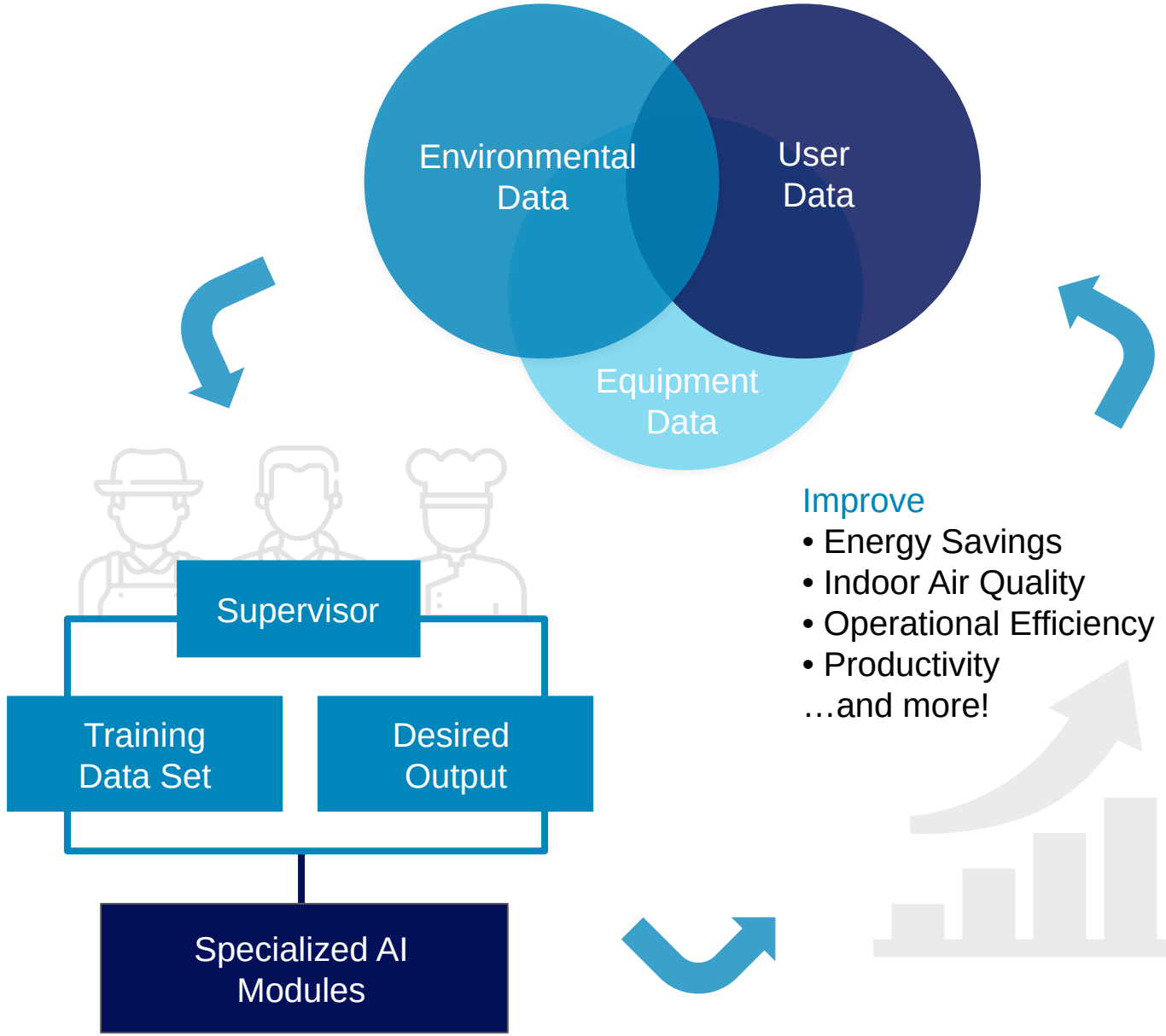


AMBIENT SYSTEMS

Climate At Your Service

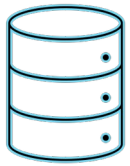
www.ambientsystems.ai

Building Optimization beyond Energy Savings



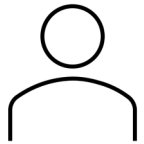
B2B SaaS Model

AMBIENT SYSTEMS



Databases

MySQL
PostgreSQL
Microsoft SQL Server
SQLite
mongoDB
redis
MariaDB
ORACLE DATABASE
Firebase
elasticsearch
amazon DynamoDB
cassandra
IBM DB2
Excel



One-time Implementation Cost

Platform Subscription

Number of Data Feeds

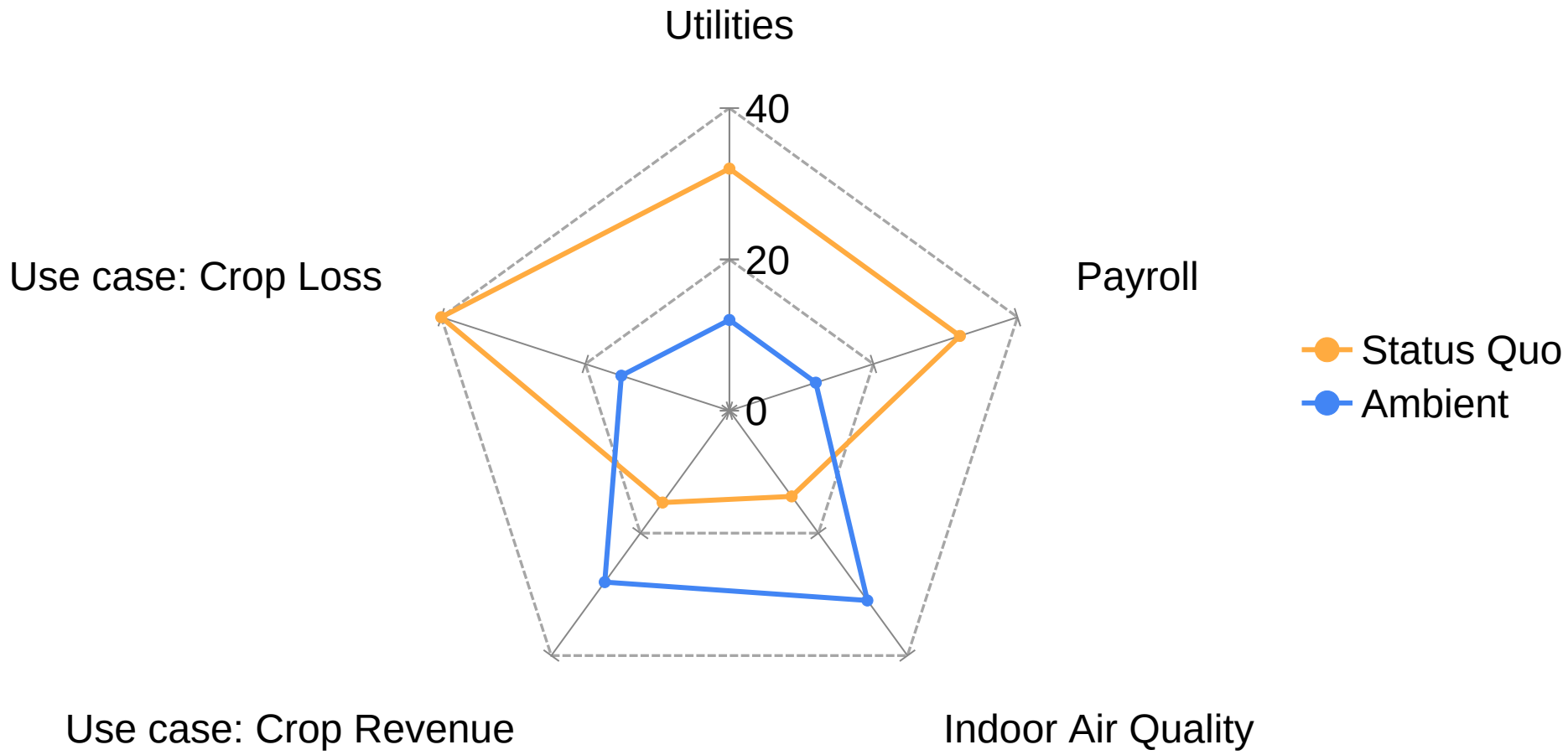
Number of AI Modules



AI Modules

- Process Optimization
- Predictive Maintenance
- Asset Optimization
- AI-based Demand Response
- AI-based Light Control

Problem: Multi-objective Optimization



Solution Targets Quantified

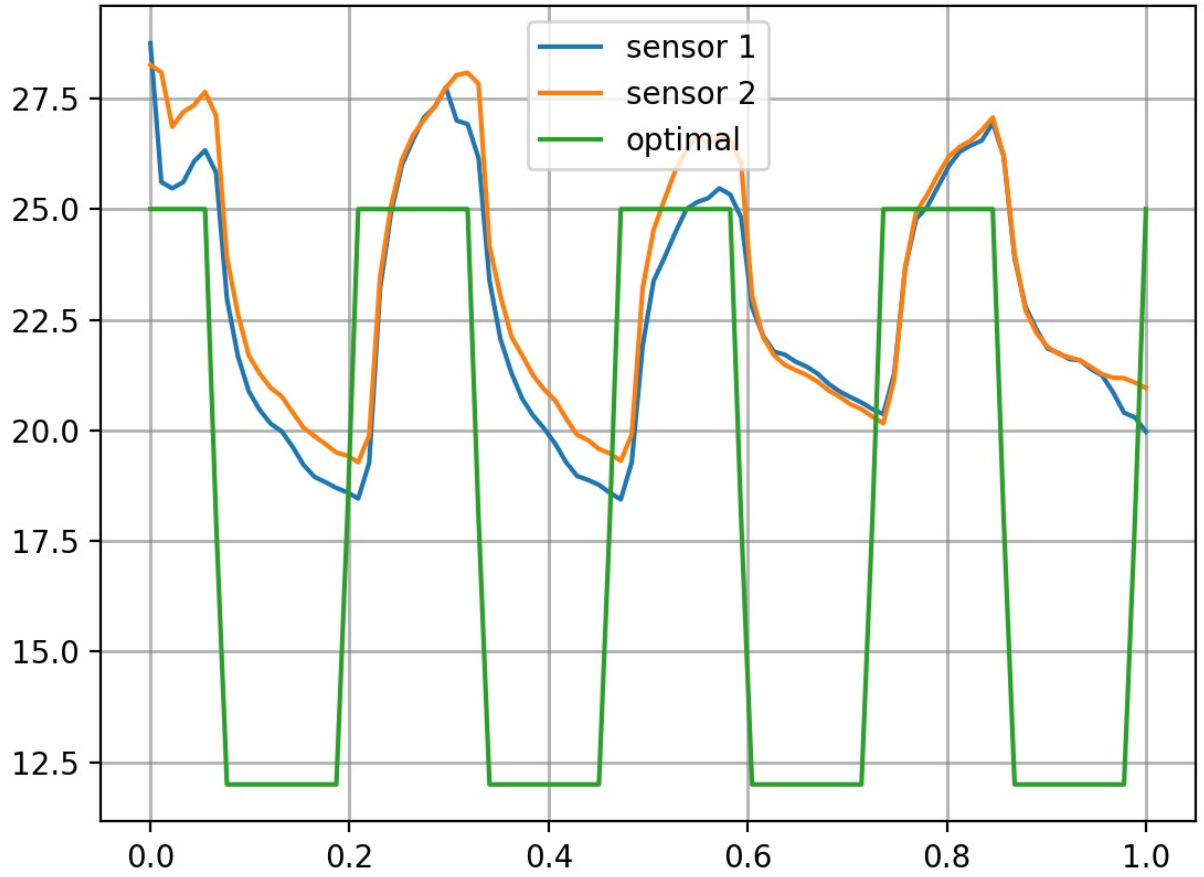
	Parameter	Actual	Optimal	Sample Action
Current POC	Air Temperature	28 C / 20 C	25 C / 12 C	Decrease temperature setpoint Increase ventilation
	Relative Humidity	55% to 70%	60% to 75%	Install fogging machines
	CO2	300 to 700 ppm	500 to 1200 ppm (min >330 ppm)	Decrease/increase ventilation (depends on CO2 concentrations outside) Install CO2 regulator
	Spectrum Light	2300 lux	5000+ lux	Turn off non-farming lights Reduce light intensity

Solution Targets Quantified

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	Spectrum Light	2300 lux	5000+ lux	Turn off non-farming lights Reduce light intensity
Full-scale Plot	Soil Temperature	MVP 2.0		
	Soil Moisture			
	pH level			
	Electrical conductivity (EC_w)			
	Crop specific parameters			

Temperature

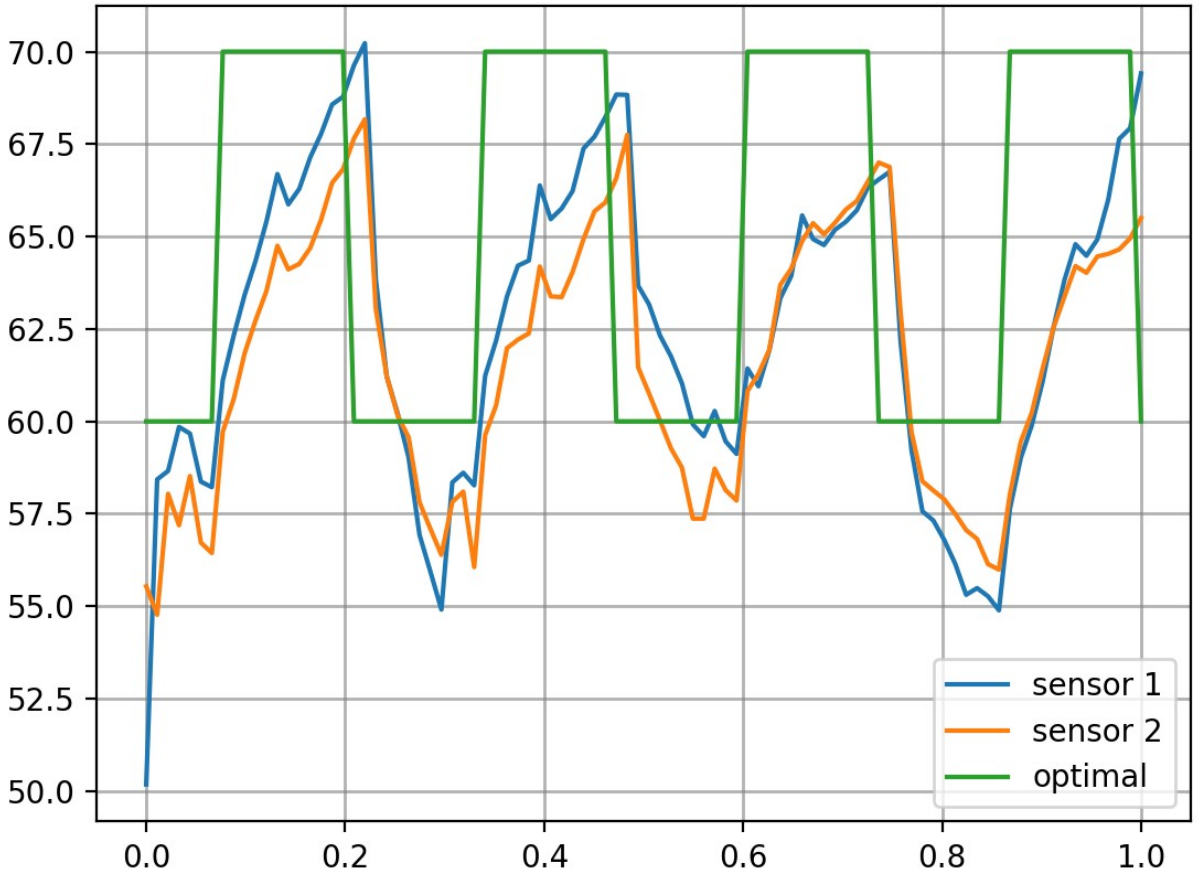
Leaf and petiole growth
Brighter leaf color
Greater sucrose
More biomass at the roots



Temperatures after bloom affect plant growth and fruit quality of strawberry, , 2000

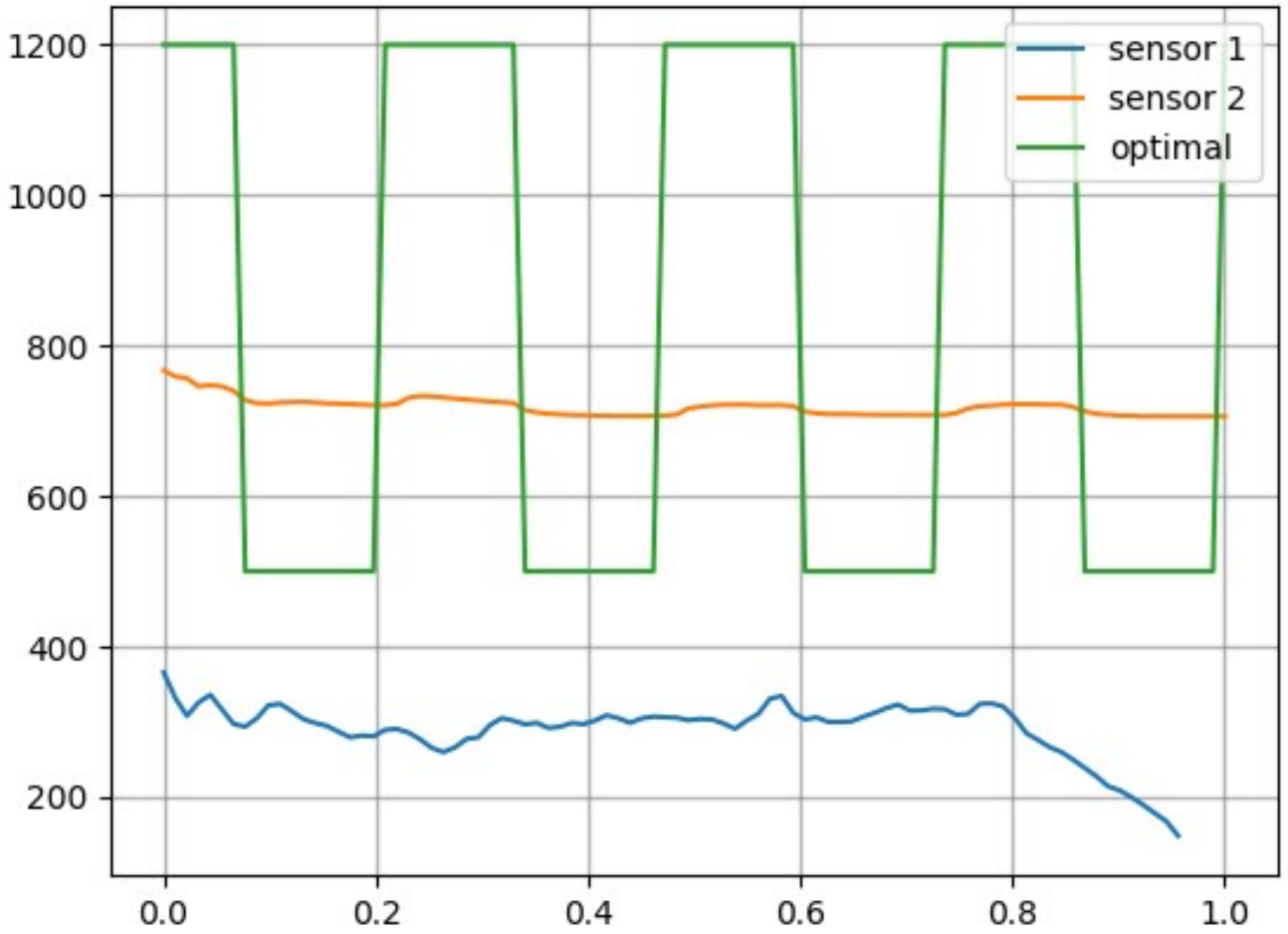
Relative Humidity

Avoid tip-burn and calyx burn
Avoid fungi
Avoid powdery mildew
Avoid leaf spot (bacterial blight)



Crop Guide: Growing Strawberry, , 2004

CO2

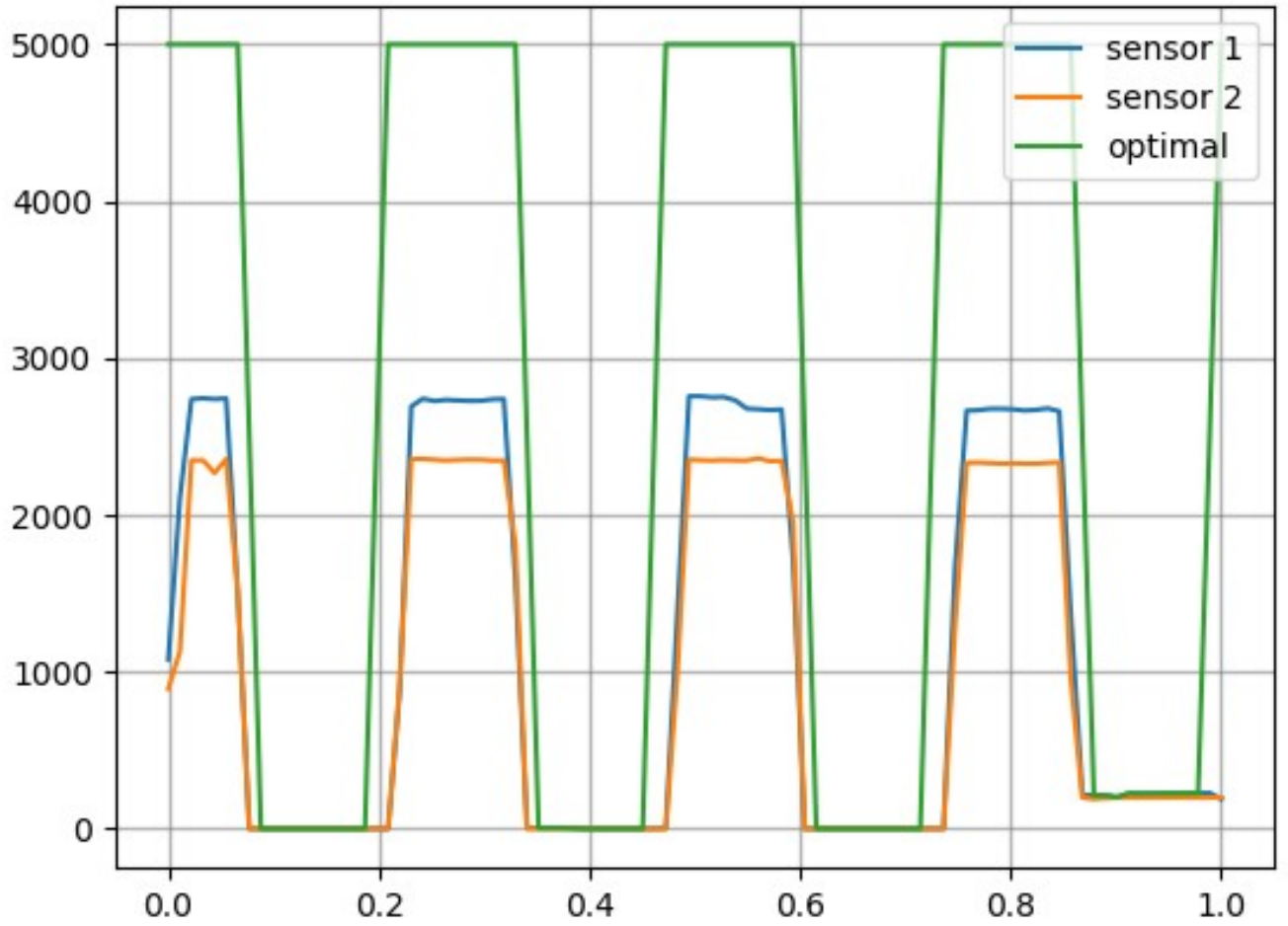


Increased photosynthesis
Earlier flowering
Higher fruit yields
Improved stem strength and flower size

Carbon Dioxide In Greenhouses, Ministry of Agriculture, Food and Rural Affairs, 2021
Carbon dioxide (CO2) applications in indoor growing,

Light Intensity

Increased photosynthesis

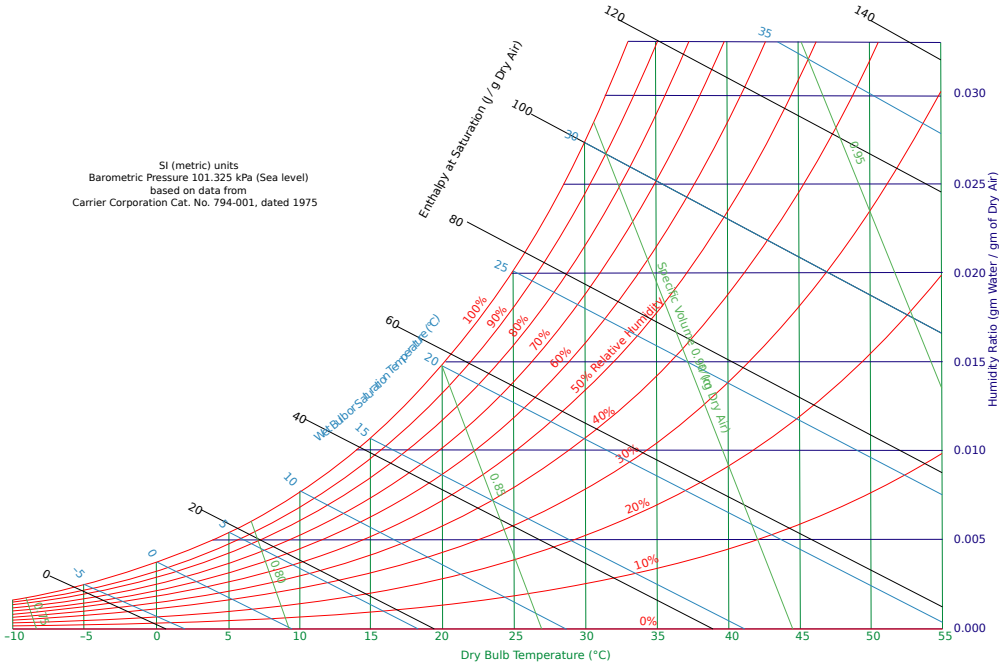


Effect Of Light Intensity, Temperature And Co2 Concentration On Photosynthesis And Growth Of In Vitro, International Society for Horticultural Science, 1989

Why not just change the AC settings?

#1: Interactions

- Psychometrics: temperature and relative humidity
- Light increases temperature and decreases humidity
- Decreasing temperature decreases CO2 concentrations in the air
- Increased ventilation introduces hot humid air from outside that needs to be cooled, purified, e hed, and re-humidified → very energy intensive → very expensive
- Safety: CO2 is needed by the plants, but high CO2 is toxic or lethal for human workers
- Photosynthesis process: lights and CO2 interact (both need to be high or both need to be low)
- Infiltration/exfiltration: porous building materials



#2: Equipment



- Building equipment is often outdated
- AC controls are limited: only allows setting temperature and fan speed (not granular enough)
- AC controls cannot be managed remotely
- Need additional or custom equipment based on plant type, growing stage, and outdoor climate

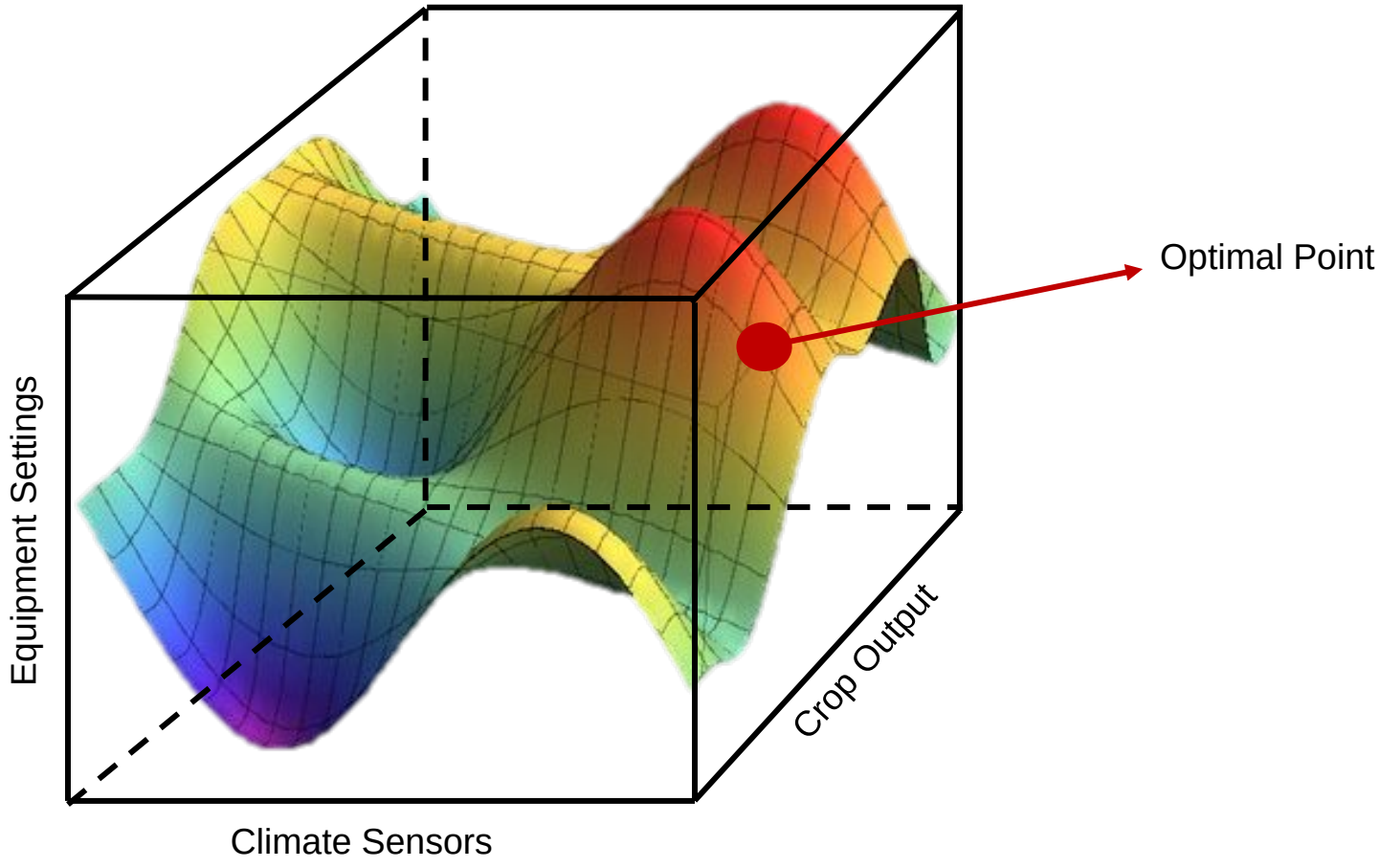




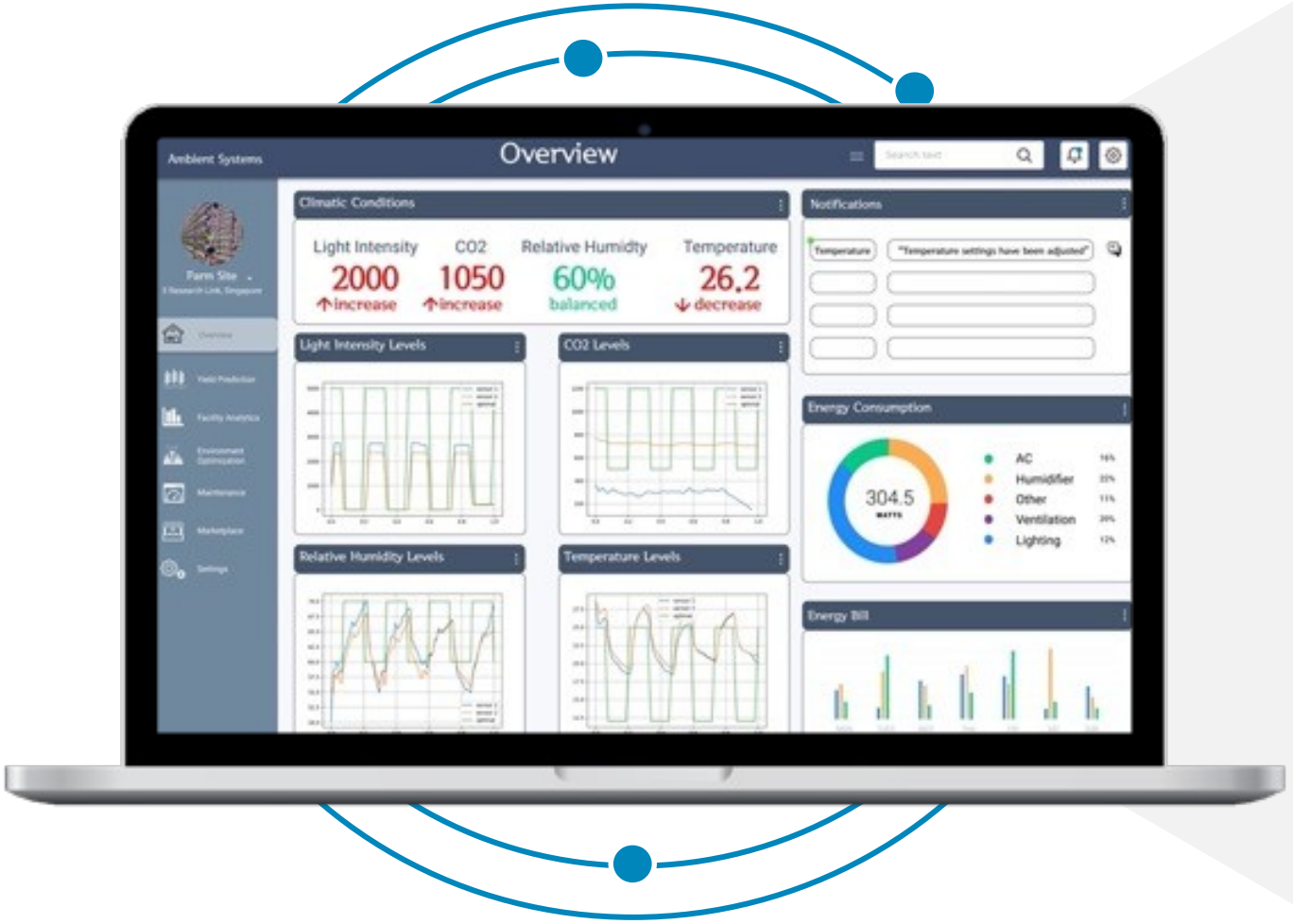
How to address this?

Optimization using building science-based AI models

Optimization: Multi-Dimensional Gradient Descent



AI Platform Backed by IP





Sense



Optimize

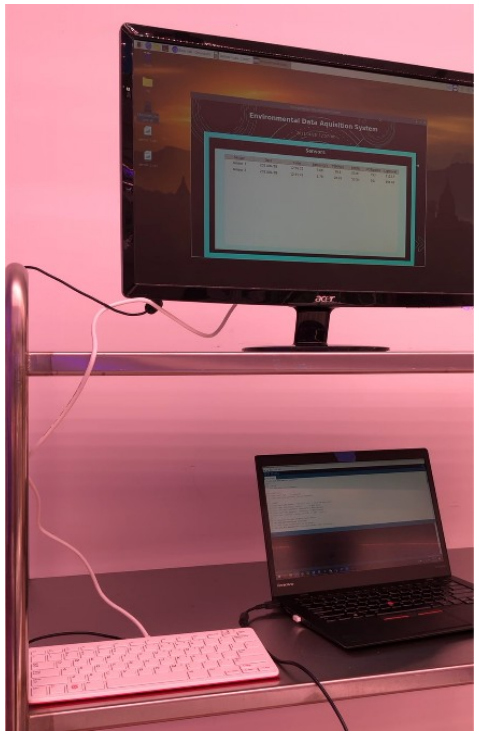


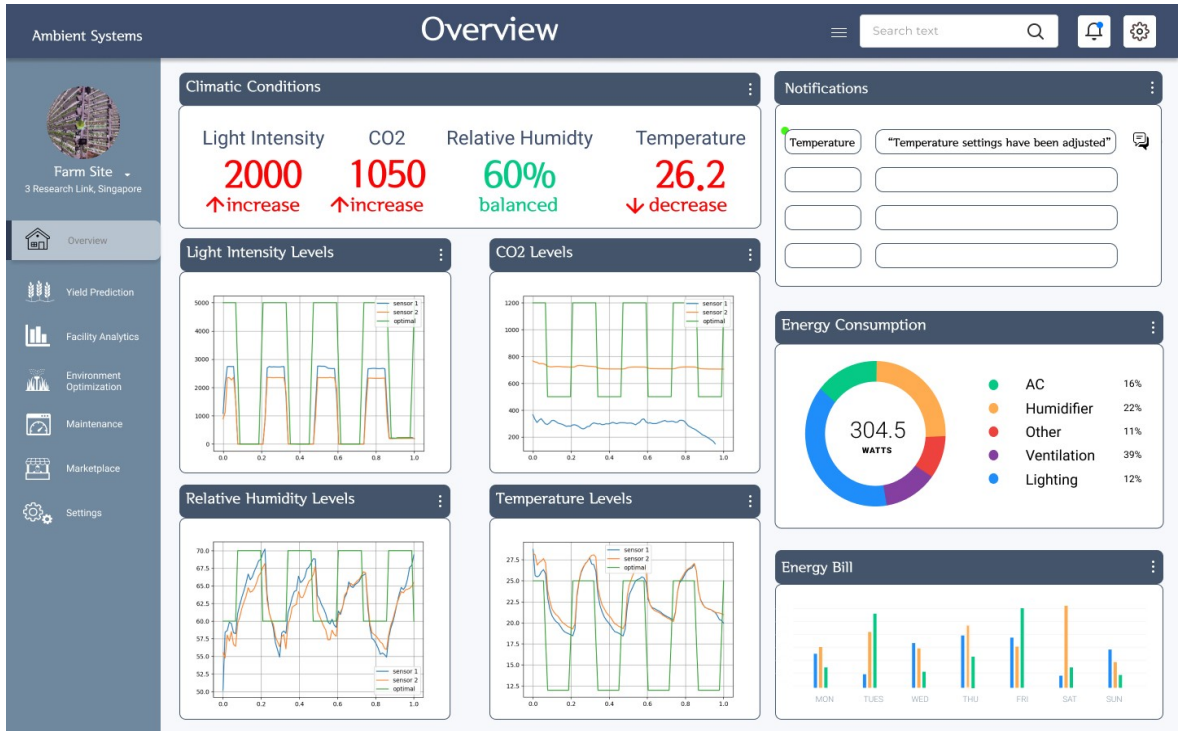
Control

Patent Pending
Invention from
National University of
Singapore (NUS)

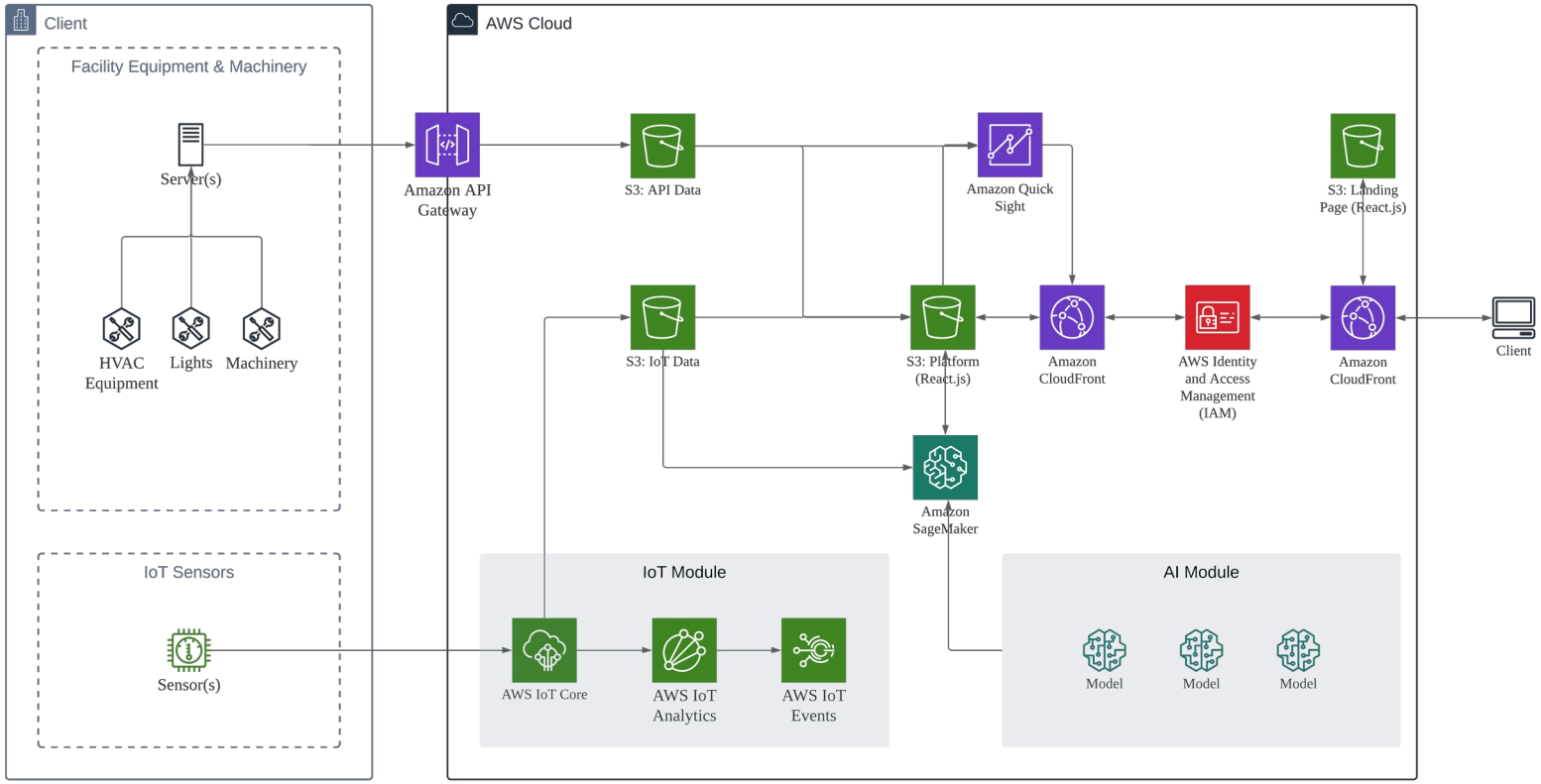
MVP v01

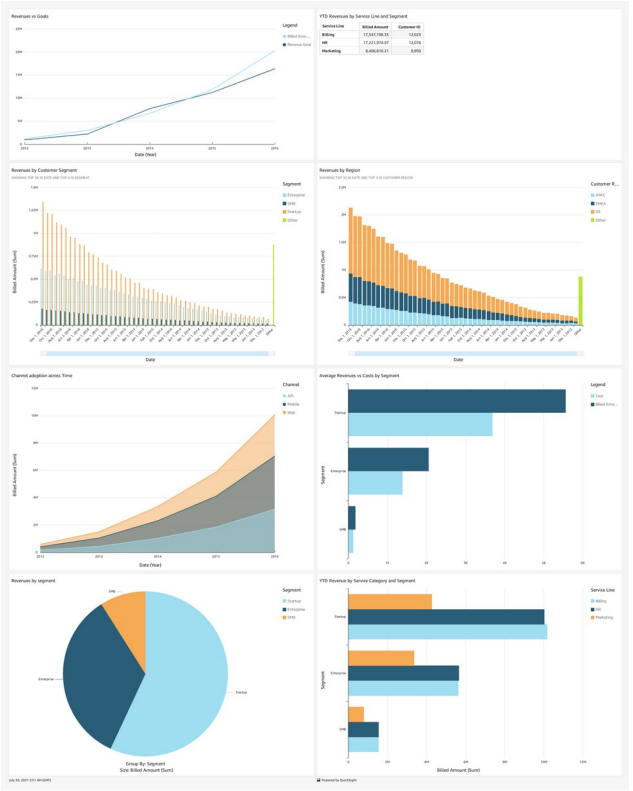

sensor 2



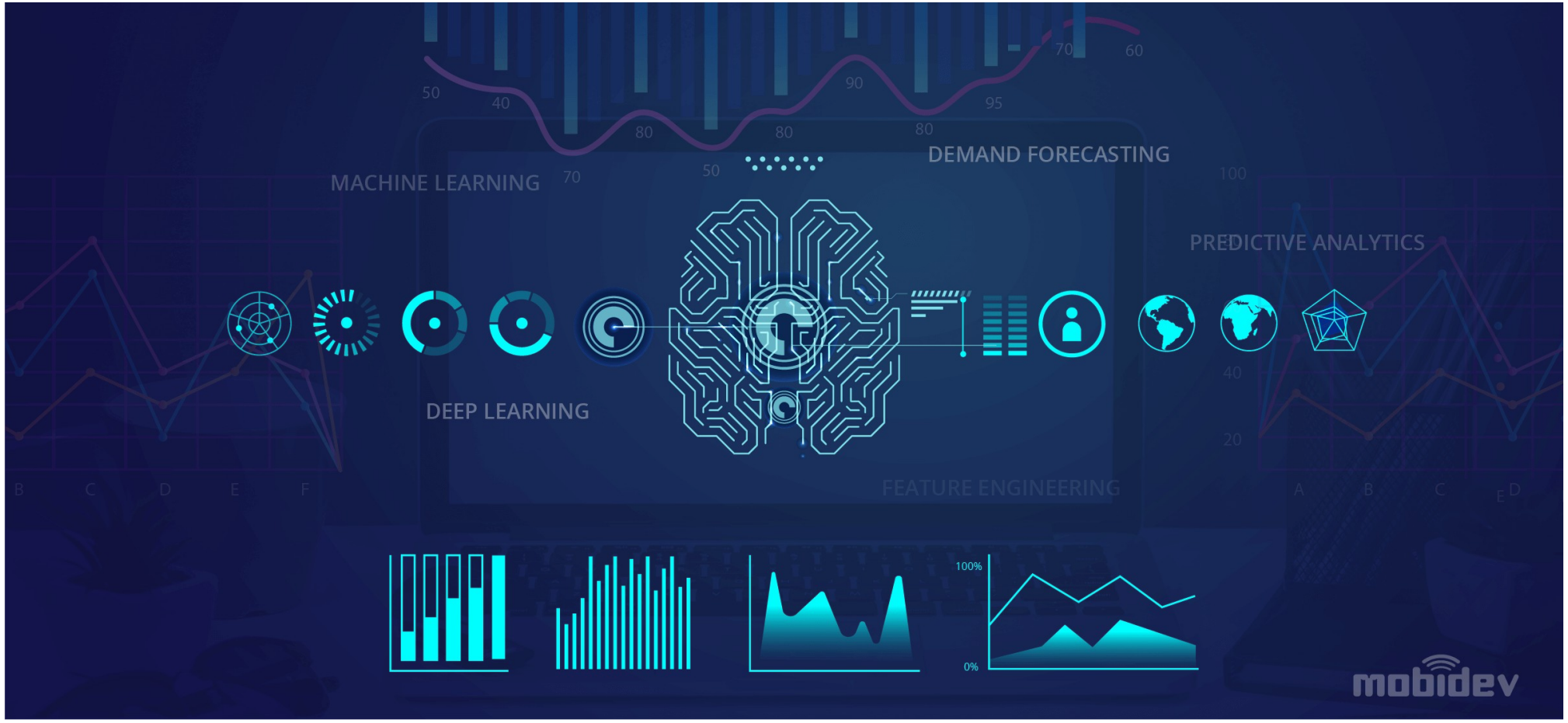

Ambient Systems
Overview
Search text
Farm Site
3 Research Link, Singapore
Overview
Yield Prediction
Facility Analytics
Environment Optimization
Maintenance
Marketplace
Settings
Climatic Conditions
Light Intensity
2000
increase
CO2
1050
increase
Relative Humidty
60%
balanced
Temperature
26.2
decrease
Notifications
Temperature
"Temperature settings have been adjusted"
Light Intensity Levels
CO2 Levels
Relative Humidity Levels
Temperature Levels
Energy Consumption
304.5
WATTS
AC 16%
Humidifier 22%
Other 11%
Ventilation 39%
Lighting 12%
Energy Bill
MON
TUES
WED
THU
FRI
SAT
SUN

MVP v02


Client
Facility Equipment & Machinery
Server(s)
HVAC Equipment
Lights
Machinery
IoT Sensors
Sensor(s)
AWS Cloud
Amazon API Gateway
S3: API Data
Amazon Quick Sight
S3: Landing Page (React.js)
S3: IoT Data
S3: Platform (React.js)
Amazon CloudFront
AWS Identity and Access Management (IAM)
Amazon CloudFront
Client
Amazon SageMaker
IoT Module
AWS IoT Core
AWS IoT Analytics
AWS IoT Events
AI Module
Model
Model
Model



MVP v03


MACHINE LEARNING
DEMAND FORECASTING
PREDICTIVE ANALYTICS
DEEP LEARNING
FEATURE ENGINEERING
100%
0%
mobidev

Data Sources

Neural Network

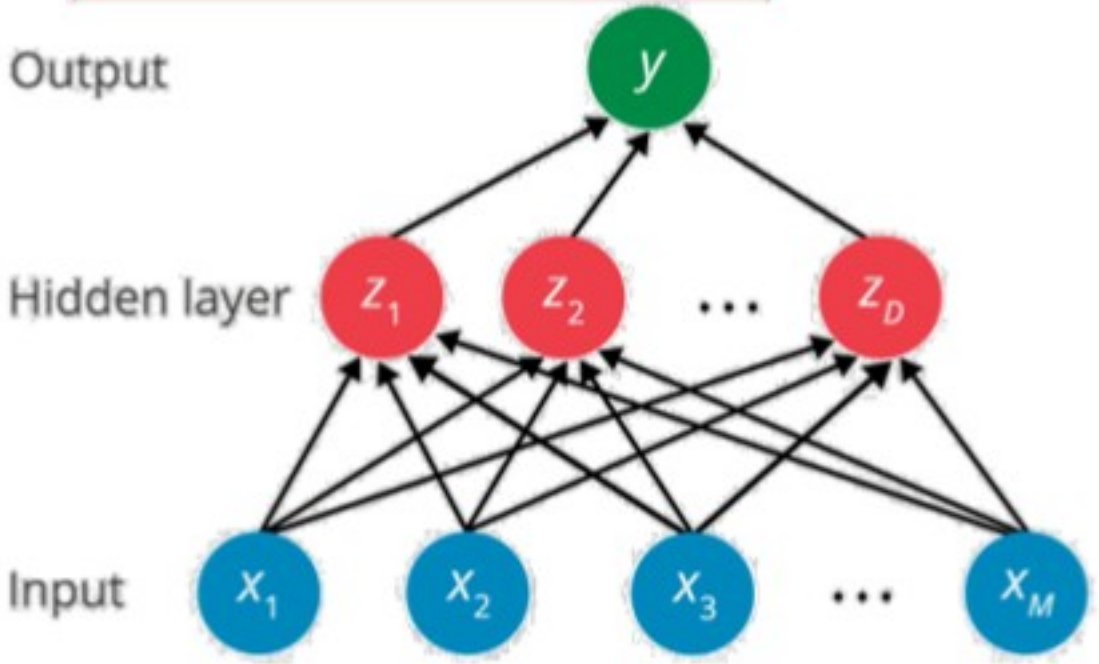
(D) Output (linear): $b = \sum_{j=0}^{D} \beta_j z_j$

(C) Hidden (sigmoid): $z_j = \frac{1}{1 + \exp(-a_j)}, \forall j$

(B) Hidden (linear): $a_j = \sum_{i=0}^{M} \alpha_{ji} x_i, \forall j$

(A) Input: Given $x_i, \forall i$

Allows to get any positive or negative real number



Optimization Function

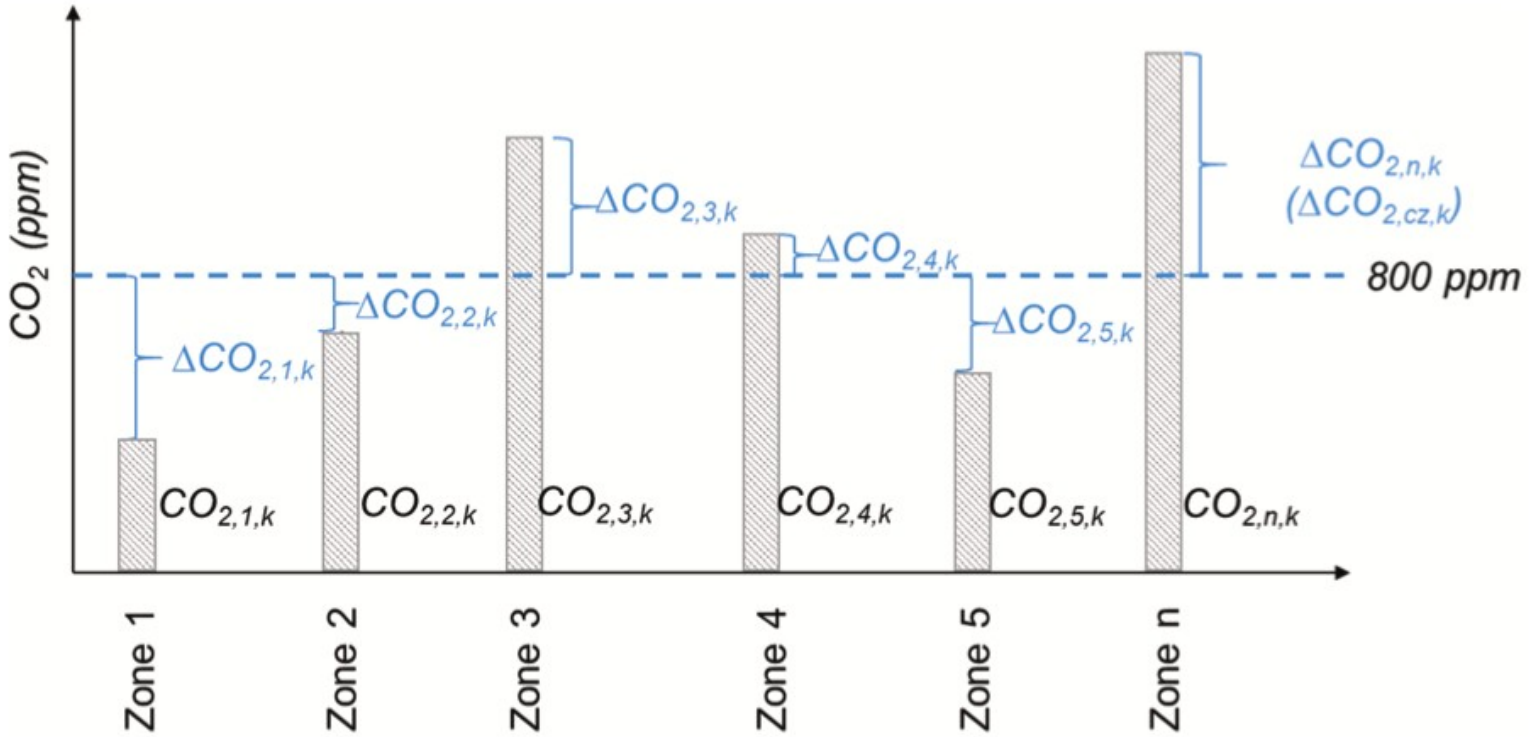
- Optimization Objectives (O):
  - Utilities: Measured by Energy Consumption ($)
  - Indoor Air Quality: Measured by CO2
  - Productivity: Measured by Crop Values ($)

$$Objective = \alpha_{Utilities} * O_{Utilities} + \alpha_{IAQ} * O_{IAQ} + \alpha_{Productivity} * O_{Productivity}$$

Utilities (Electricity)

- Sample equipment (i):
  - Chiller Plant: provides cooling to all the zones
  - AHU Fan: Delivering Airflow to the zones
  - Auxiliary Devices: Fans, AC
- Sum the energy consumption (V*A) across time series (t: 0 to time interval) and relevant equipment (j: 1 to # of equipment)

IAQ



$$Obj_{IAQ} = \sum_{i=1}^{n} \max\{0, (CO2_i - TH_{CO2})\}$$

Productivity

- Need to work with AgTech experts to identify appropriate function for crop productivity
- Productivity of the farm can be calculated based on following variables:
  - Y: Crop Yield (kg / m2)
  - V: Crop Value ($ / kg)
- Summed across:
  - Crops (i): strawberries, lettuce
  - Harvest Tier (j): premium strawberries, standard strawberries, spoiled (e.g., due to mold or pests)
  - Time (t): harvest cycle (e.g., hours, days, weeks, months)

Optimization Process

- Three steps:
  - Establish multi-agent system (define model)
  - Reformulated to multiple independent sub-problems
  - Iteration
- General: determine local optimal then global optimal
- Multiple control strategies:
  - #1: Fixed Setting
  - #2: User Adjusts
  - #3: Optimum Calculated by ML then Set
  - #4: Real-time calculations & adjustments

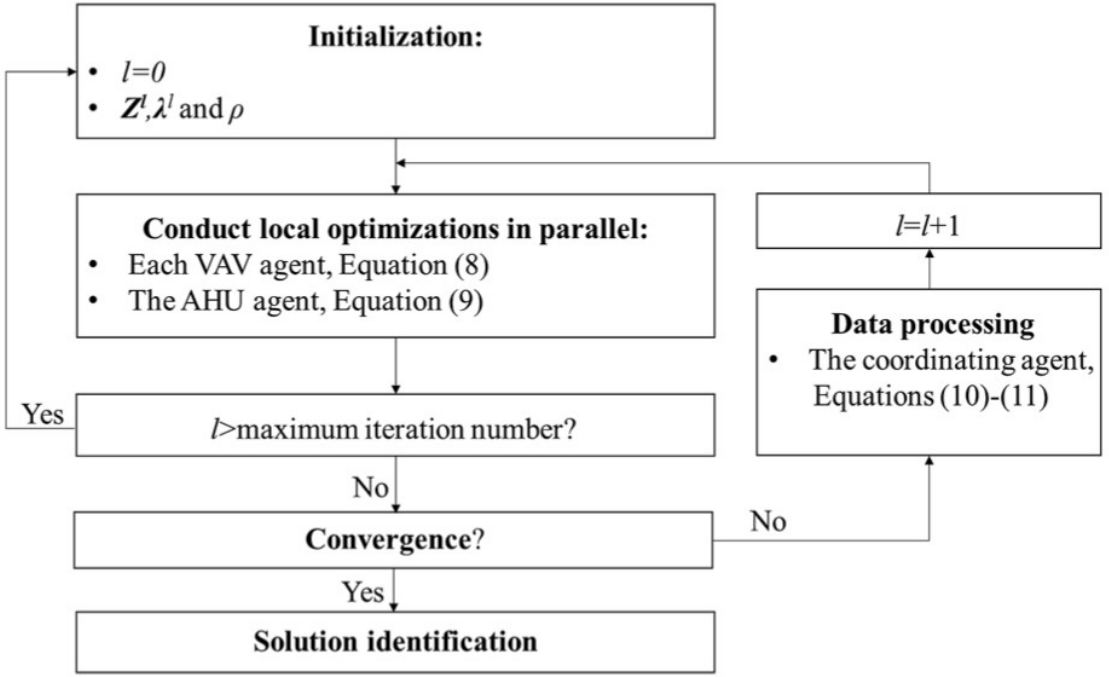


Fig. 6. Flowchart for the solution identification using ADMM.

Datasets: Simulation

Variable	Unit	Type	Distribution	Parameters
Equipment Setting				

Variable	**Unit**	**Type**	**Distribution**	**Parameters**
Inter-Arrival	sec	Continuous	$-0.001 + Exponential(\lambda)$	$\lambda = 1.35$
Base Station	units	Discrete	$DiscreteUniform(N)$	$N = 19$
Call Duration	sec	Continuous	$Exponential(\theta)$	$\theta = 119$
Location Between Base Stations	km	Continuous	$Uniform(\alpha, \beta)$	$\alpha = 0$ $\beta = 2$
Car Speed	km/hr	Continuous	$Triangular(a, b, c)$	$a = 70$ $b = 90$ $c = 110$

Table 1: Summary of Input Variable Distributions

Patent Classification

A	HUMAN NECESSITIES
B	PERFORMING OPERATIONS; TRANSPORTING
C	CHEMISTRY; METALLURGY
D	TEXTILES; PAPER
E	FIXED CONSTRUCTIONS
F	MECHANICAL ENGINEERING; LIGHTING; HEATING; WEAPONS; BLASTING
G	**PHYSICS** **G06:** COMPUTING; CALCULATING OR COUNTING **G06N:** COMPUTER SYSTEMS BASED ON SPECIFIC COMPUTATIONAL MODELS **G06N 3/00**: Computer systems based on biological models **G06N 5/00**: Computer systems using knowledge-based models **G06N 7/00**: Computer systems based on specific mathematical models **G06N 20/00**: Machine learning
H	ELECTRICITY

POC Pilot Outcomes

Recommendations

Insight	Recommendation	Cost
No historical data	Measure and securely archive critical operational data for crop repeatability and insurance	Ambient Platform Tier 1
Variability between days	Monitor for longer period of time (entire harvest cycle)	Ambient Platform Tier 1
No utilities data	Monitor energy consumption	$5k-$10k hardware modification Ambient Platform Tier 1 & 2
High temperature	Decrease daytime and night-time temperature	$5k - $10k portable AC
Low CO2	CO2 regulator (increase CO2)	$1k - $2k CO2 regulator $500 monthly CO2 supply
Moisture timing	Introduce mist earlier in the day Reduce mist later in the day	Ambient Platform Tier 1
Plants have various optimal parameter schedules	Determine farm portfolio optimal (not just for one plant but entire farm)	Ambient Platform Tier 1 & 3

Thank You

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